

Fine-Structure Constant from Spectral Volume of the D^* -Sphere

David A. Besemer^{1,*}

¹*Independent Researcher*

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We derive the inverse fine-structure constant α_{em}^{-1} as the volume of a D^* -dimensional sphere, corrected by the elementary lattice tension. Using the exact tree-level and fixed-point values of D^* , we find $\alpha_{em}^{-1} \approx 136.96$, in excellent agreement with the high-energy value of the coupling constant at the Planck scale.

I. INTRODUCTION

We calculate the electromagnetic coupling constant at the Planck scale. The formulation uses the geometric volume of a sphere in fractional dimensions.

II. THEORETICAL FORMULATION

Here we formulate the core mathematical relations governing the system:

$$\alpha_{em}^{-1} = 2\pi(D^*)^3 + \delta D^* \quad (1)$$

$$\alpha_{em}^{-1}(D_{tree}^*) = 2\pi\left(\frac{67}{24}\right)^3 + \frac{1}{24}\left(\frac{67}{24}\right) \approx 136.955 \quad (2)$$

III. GEOMETRIC VOLUME CALCULATION

The formula combines the spatial volume $2\pi(D^*)^3$ with a boundary tension term δD^* , representing the energy density of the lattice boundary.

IV. CONCLUSION

Electromagnetism emerges as a boundary effect of the fractal spacetime manifold, with a coupling fixed by the spectral dimension.

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- [3] D. A. Besemer, `sdgft` Python package, <https://github.com/david-besemer/sdgft> (2026).

* david.besemer@sdgft.org